

Modern Data Analysis in Economics

Martin J. Conyon

Bentley University

Required readings

There is **no Mankiw chapter** for this material. It is built from published research, and these are the papers it works through.

- **Card, D. and A. B. Krueger** (1994), “Minimum Wages and Employment,” *American Economic Review* 84(4) — and the 1st exchange with **Neumark and Wascher** (2000), *AER* 90(5).
- **Blau, F. D. and L. M. Kahn** (2017), “The Gender Wage Gap,” *Journal of Economic Literature* 55(3).
- **Kleven, H., C. Landais and J. E. Sørensen** (2019), “Children and Gender Inequality,” *AEJ: Applied* 11(4).

Read the Card–Krueger paper first. The rest of the session is built on its design.

Where we are going

1. **Correlation is not causation** — and the three ways it goes wrong.
2. **The missing counterfactual** — the problem underneath everything.
3. **Control groups** — what a good one is, and the data economists have.
4. **Difference-in-differences** — a causal estimate from four averages.
5. **Parallel trends** — the assumption, and how the method fails.
6. **Card and Krueger** — the famous minimum-wage natural experiment.
7. **The gender wage gap** — a hard question without a ready-made design.
8. **The child penalty** — the event that supplies the missing design.

PART 1

Correlation Is Not Causation

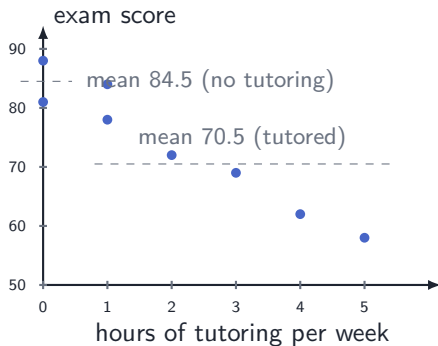
A pattern in data

Eight students in a course; hours of weekly tutoring and final exam score. (Invented data — the numbers are made up to make a point.)

student	A	B	C	D	E	F	G	H
tutoring (hrs/week)	0	0	1	1	2	3	4	5
exam score	88	81	78	84	72	69	62	58

Read along the rows: as tutoring hours rise, scores tend to **fall**.

Draw the graph



Correlation. Two variables moving together — here *negative*: more tutoring goes with lower scores.

The tempting sentence

Average score without tutoring: $(88 + 81)/2 = \mathbf{84.5}$.

Average score with tutoring: $(78 + 84 + 72 + 69 + 62 + 58)/6 = 423/6 = \mathbf{70.5}$.

$$84.5 - 70.5 = 14.0 \text{ points}$$

The tempting sentence: “*tutoring **lowers** your exam score by 14 points.*”

The correlation is a fact. The sentence is a causal claim.

They are not the same thing.

Reason 1: reverse causality

Reverse causality. The data show X and Y moving together, but the influence runs from Y to X , not from X to Y .

Cities with more police officers per resident tend to have **more** crime, not less.

Does hiring police cause crime? **No: crime causes hiring.** City councils respond to a crime wave by putting more officers on the street.

The correlation is real. The arrow points backward.

Reason 2: omitted variables

Omitted variable. A third variable, left out of the comparison, that drives both of the variables you are looking at.

Among schoolchildren, shoe size and reading ability are strongly **positively** correlated.

Big feet do not help you read. **Age** drives both: older children have bigger feet *and* read better.

X and Y move together because Z moves them both.

Reason 3: selection

Selection. The people who receive a treatment are different from those who do not — *before* the treatment does anything.

Back to the tutoring data. Ask the question the table hides: **who signs up for tutoring?**

Students who are *already struggling*. The tutored group was headed for lower scores before the first session met.

The 14-point gap mixes the effect of tutoring with the difference in who chose it. The comparison cannot separate them.

Three doors into the same room

failure	the arrow	example
reverse causality	$Y \rightarrow X$	police and crime
omitted variable	$Z \rightarrow X$ and $Z \rightarrow Y$	shoe size and reading
selection	who gets X differs	tutoring and scores

All three are the **same disease**: the group with X and the group without it **differ for reasons other than X** .

So the difference in outcomes is not the effect of X .

Active Learning 1

For each claim, **name the failure** — reverse causality, omitted variable, or selection — and **write the one-sentence story** that produces the correlation without the causal claim being true. **The story is the marks.**

- (a) Months with high umbrella sales have more rain — so umbrella sales cause rain.
- (b) Students who attend the optional review session score higher — so the session raises scores.
- (c) Towns with more churches have more bars — so churches drive people to drink.

Four minutes. Paper and pencil.

PART 2

The Missing Counterfactual

What a causal claim actually says

“Tutoring lowered Maya’s score” means:

her score **with** tutoring $<$ the score she **would have gotten** without it.

Counterfactual. What *would have happened* to the same person, firm, or town if the treatment had not occurred.

Every causal claim is a comparison between an outcome that happened and one that **did not**.

The fundamental problem

	score <i>with</i> tutoring	score <i>without</i> tutoring
Maya (tutored)	64	?
Daniel (not tutored)	?	81

Each person is observed in **one** column — never both. The cell you need is the cell you can never fill.

The counterfactual is, by definition, unobservable.
This is the fundamental problem of causal inference.

Bad substitute 1: before versus after

A town runs a summer job-training course. (Invented numbers.) Participants' average earnings:

year before: \$28,000 year after: \$31,000

Did the training add $31,000 - 28,000 = \$3,000$?

Only if **nothing else changed** in a year: no recovery, no inflation, no raises that were coming anyway.

“Before” is a bad stand-in for “without”: time moved everything at once.

Bad substitute 2: treated versus untreated

The second reflex: compare the tutored with the untutored.

$$84.5 - 70.5 = 14.0 \text{ points}$$

We have already seen the flaw: the groups **differed before the treatment** — the struggling students selected in.

“The others” is a bad stand-in for “without”: the groups were never alike.

The question that organizes everything

Every empirical method in economics is an answer to one question:

Where does your counterfactual come from?

- Before vs. after — *from the past*. Fails when time matters.
- Treated vs. untreated — *from other people*. Fails when groups differ.
- The methods ahead **combine** the two so each failure cancels the other.

Active Learning 2

A hospital offers knee-surgery patients an optional rehab program. Its brochure reports:
“patients who enrolled walked unaided in 6 weeks; patients who declined took 9.”

- (a) **Before any arithmetic:** is the brochure's implied effect likely **overstated or understated**? Predict the direction, then give the reason. **The reason is the marks.**
- (b) Which of the three failures from Part 1 is this?
- (c) What comparison would you *want* instead? One sentence.

Three minutes. Paper and pencil.

PART 3

Control Groups and the Data We Have

What a good control group is

Control group. Units that do not receive the treatment, whose outcome stands in for what the treated group's outcome *would have been*.

The job description is strict. A good control group is

the same as the treated group in every way that matters — except the treatment.

Then any gap in outcomes has only one possible source left.

Randomization: the ideal

Randomized controlled trial. Assign the treatment by pure chance — a coin flip decides who is treated and who is control.

Why chance? With enough subjects, the coin **balances the two groups on everything**:

- on what you can measure — age, income, grades;
- and on what you **cannot** — motivation, health, talent.

Nobody can select in. The arrow cannot run backward.
Chance has no favorites.

Why economists usually cannot randomize

Try to design the trial:

- flip a coin for which **states** get a higher minimum wage;
- assign **recessions** to a random half of the country;
- randomize who is born **male or female**.

Impossible, unaffordable, or unconscionable — usually all three.

Economists mostly cannot run the ideal experiment.
We must find its accidental substitutes.

The data we actually have

Experimental data. Generated by a researcher who controlled the assignment — who was treated, who was not.

Observational data. Generated by the world going about its business: surveys, tax records, censuses, payrolls.

Nearly all economic data is **observational** — nobody assigned anything. Every failure from Part 1 is therefore **on the table by default**.

Three shapes of data

shape	what it holds	example
cross-section	many units, one date	50 states, this year
time series	one unit, many dates	one state, 30 years
panel	many units, many dates	50 states, 30 years

A cross-section can compare groups *now*. A time series can compare *before and after*.
A **panel can do both at once** — and that is what the method ahead requires.

A panel, waiting for a question

Two neighboring towns; vacant storefronts on each Main Street, counted every December. (Invented data.)

	Dec 2023	Dec 2024	Dec 2025
Fairview	27	24	18
Grantham	33	30	27

Two units, three dates: a small **panel**. In January 2025, something happened in Fairview — and only in Fairview.

Natural experiment. An event — a law, a border, a storm — that hands a treatment to one group and not another, in a way unrelated to how the groups were already doing.

Nobody flipped a coin, but the world came close: the event, not the outcome, decided who was treated.

The craft: recognize the accident, then find the control group it created.

Active Learning 3

Your university will scrap its math placement exam next fall. The registrar proposes to measure the effect on first-year GPA: *“compare next year’s first-years (no exam) with this year’s (exam).”*

- (a) What is the proposed **control group**, and what counterfactual is it standing in for?
- (b) Give **two distinct reasons** it is a poor stand-in. **The reasons are the marks.**
- (c) Suggest one comparison that would do better. One sentence.

Four minutes. Paper and pencil.

PART 4

Difference-in-Differences

The setup

January 2025: **Fairview** begins paying a \$500 grant to any landlord who re-lets an empty storefront. **Grantham**, one county over, does not. (Invented policy, invented numbers.)

The question: **did the grant reduce vacancies?**

The threats:

- both towns ride the **same regional economy** — a good year cuts vacancies everywhere, grant or no grant;
- the towns **differ permanently** — Grantham's Main Street has always been emptier.

Plug the numbers in

vacant storefronts	Dec 2024 (before)	Dec 2025 (after)
Fairview (grant)	24	18
Grantham (no grant)	30	27

Four numbers: **two groups** $\times$ **two dates**.

Everything the method needs is in this 2×2 table.

Subtract once: the change in each town

	before	after	change
Fairview (grant)	24	18	$18 - 24 = -6$
Grantham (no grant)	30	27	$27 - 30 = -3$

Fairview's $-6 = \text{grant} + \text{shared year}$.

Grantham's $-3 = \text{shared year alone}$.

Comparing each town *with itself* has already removed the permanent Fairview–Grantham gap.

Subtract twice: the estimate

$$\underbrace{(-6)}_{\text{grant+year}} - \underbrace{(-3)}_{\text{year}} = -3$$

The shared year cancels. What remains is the grant: **about 3 storefronts re-let because of the policy.**

Difference-in-differences (DiD). The change in the treated group minus the change in the control group: an estimate of the treatment's effect, built from four averages.

The check: subtract the other way

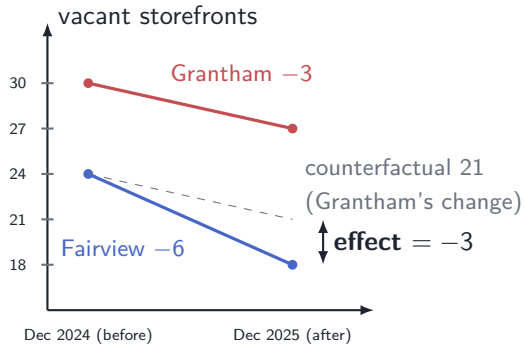
Same table, other direction — compare the *towns* first, then the *dates*:

$$\text{before: } 24 - 30 = -6 \quad \text{after: } 18 - 27 = -9$$

$$(-9) - (-6) = -3$$

Rows first or columns first — the answer must agree.
If your two routes disagree, go hunting for the error.

Draw the graph



The recipe

1. Find a **treated group** and a **control group** the same event did not touch.
2. Measure the outcome in both, **before and after** — four averages.
3. Subtract along time: each group's **change**. The permanent group gap drops out.
4. Subtract across groups: **change minus change**. The shared trend drops out.

What survives both subtractions is the estimate of the effect.

What the design needs

The subtraction treated Grantham's -3 as **what would have happened in Fairview** without the grant.

That is an **assumption**, not a fact. It has a name — and it can fail.

DiD is only as good as the claim that the control's change reveals the treated group's counterfactual change.

Active Learning 4

Elm branch library opens on Sundays starting July; Oak branch does not. Weekly visits (invented):

	June (before)	September (after)
Elm (Sunday hours)	480	560
Oak (unchanged)	400	430

- (a) Before computing: what job is **Oak** doing in this design? One sentence.
- (b) Compute the DiD estimate **both ways** — rows first, then columns first. **Both routes are the marks.**

Four minutes. Paper and pencil.

PART 5

Parallel Trends — and How the Method Fails

The assumption, stated plainly

Parallel trends. Without the treatment, the treated group's outcome would have changed by **the same amount** as the control group's did.

Not: the groups are *at the same level*. (They need not be — levels cancel.)

But: absent the policy, the two lines would have **moved in parallel**.

It is a claim about a world that never ran.
It cannot be proven — but it can be probed.

Probing it: the pre-period

The panel had a third column — **before the policy existed**:

	Dec 2023	Dec 2024	pre-trend
Fairview	27	24	$24 - 27 = -3$
Grantham	33	30	$30 - 33 = -3$

In the year *before* the grant, the towns moved **in lockstep**: -3 and -3 .

Parallel before. The -3 estimate has earned some trust.

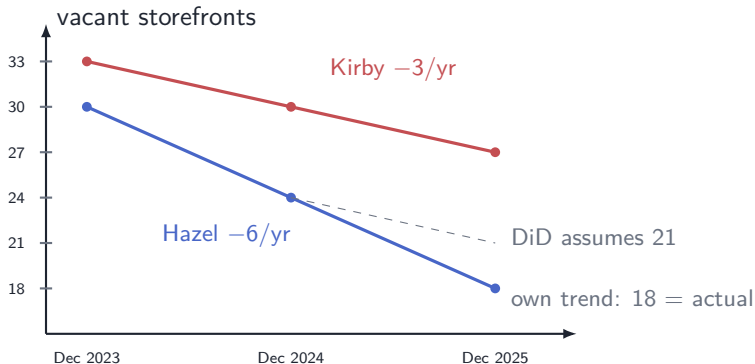
The twin case: same table, different history

Two other towns, **Hazel** (grant) and **Kirby** (no grant). Their 2×2 is **identical** — so DiD gives -3 again. But their pre-period differs:

	Dec 2023	Dec 2024	pre-trend	Dec 2025
Hazel (grant)	30	24	$24 - 30 = -6$	18
Kirby (none)	33	30	$30 - 33 = -3$	27

Before any policy, Hazel was **already improving twice as fast**. The lines were **diverging**.

The verdict: DiD gets the twin case wrong



On its own trend, Hazel reaches $24 - 6 = 18$ **with no grant at all**: true effect ≈ 0 .
DiD reports **-3** — wrong by the trend gap, $(-6) - (-3) = -3$.

Fairview–Grantham and Hazel–Kirby have **identical** 2×2 tables. Same arithmetic, same -3 .

One estimate is credible. The other is an artifact of a trend that was there all along.

The four numbers cannot tell you which case you are in.
Only the history before the policy can.

Always demand the pre-period. No pre-trend evidence, no credibility.

Active Learning 5

Alder County starts a monthly job fair in early 2025; Birch County does not. Employed residents, in hundreds (invented):

	2023	2024	2025
Alder (job fair from 2025)	100	120	145
Birch (none)	100	110	122

- (a) Compute the DiD estimate for 2025, **both routes**.
- (b) **Test the assumption** the estimate leans on.
- (c) What is your honest estimate of the job fair's effect? **The reasoning is the marks.**

Five minutes. Paper and pencil.

PART 6

Card and Krueger: The Minimum-Wage Natural Experiment

The event

April 1, 1992: New Jersey raises its minimum wage from \$4.25 to \$5.05 per hour. Pennsylvania, next door, stays at \$4.25.

The standard model's prediction, restated: a minimum wage is a **price floor** for labor. Set above the market-clearing wage, it makes labor cost more — and employers respond by hiring **fewer** hours. **Employment falls.**

A law changed on one side of a state line and not the other:
a natural experiment, ready-made.

David Card and Alan Krueger surveyed **410 fast-food restaurants** — Burger King, KFC, Wendy's, Roy Rogers — in New Jersey and **eastern Pennsylvania**:

- **before**: February–March 1992;
- **after**: November–December 1992.

Why fast food? Minimum-wage jobs, uniform product, chains on both sides of the line.
Why *eastern* PA? **Same regional economy** — the strongest available case for parallel trends.

Card and Krueger (1994), *American Economic Review* 84(4), 772–793.

Plug the numbers in

Average full-time-equivalent employment per restaurant:

	before	after	change
New Jersey (raise)	20.44	21.03	$21.03 - 20.44 = +\mathbf{0.59}$
Pennsylvania (none)	23.33	21.17	$21.17 - 23.33 = -\mathbf{2.16}$

$$\text{DiD} = 0.59 - (-2.16) = +\mathbf{2.75}$$

Check, other route: gaps $20.44 - 23.33 = -2.89$; $21.03 - 21.17 = -0.14$;
 $-0.14 - (-2.89) = +\mathbf{2.75}$.

(From the paper's published table of means; the paper reports 2.76, computed before rounding.)

What it found

Employment in New Jersey **did not fall** relative to Pennsylvania — if anything, it **rose**.

The simple price-floor prediction — employment falls — **did not appear** in this data.

What the study does *not* say:

- not “minimum wages never reduce employment”;
- not “the model is worthless” — it says: *in this episode, at this size of increase, the predicted fall is not visible.*

The dispute

Neumark and Wascher (2000): re-ran the comparison with **payroll records** collected from restaurants, instead of phone-survey answers — and found NJ employment **falling** about 4% relative to PA.

Card and Krueger (2000): replied with **government administrative data** covering nearly all establishments — and again found **no relative decline**.

Both in *American Economic Review* 90(5).

Same design. Same states. Same dates. Different *data* — different answer.

The lesson

The method is only as good as **two** things:

1. **the comparison** — is the control's trend really the treated group's counterfactual?
2. **the data** — do the four numbers measure what they claim to measure?

Thirty years on, minimum-wage studies still argue about exactly these two questions.

That is not a scandal. That is what empirical economics looks like from inside.

Active Learning 6

Card and Krueger's estimate: +2.75.

- (a) Invent **two different stories** under which the *true* effect of the raise was zero or negative, yet the design still produces +2.75. For each, say which pillar breaks: **the comparison** or **the data**. **The stories are the marks.**
- (b) Which pillar was the Neumark–Wascher critique aimed at?

Four minutes. Paper and pencil. No arithmetic — reasoning only.

PART 7

The Gender Wage Gap

The raw fact

In 2023, the median woman working full-time in the United States earned about **84 cents** for each dollar earned by the median man working full-time.

$$\text{raw gap} \approx 100 - 84 = 16\%$$

Bureau of Labor Statistics, median usual weekly earnings, full-time wage and salary workers, 2023.

The number itself is not in dispute.

The dispute — and it is fierce — is over what it *means*.

There is no Pennsylvania here

Hold this question against the designs we have built:

- You **cannot randomize sex**.
- There is **no before and after** — no date on which treatment began.
- Comparing men with women is **not treatment against control** in any sense we have constructed: the groups differ in histories, occupations, hours, constraints — everything a control group must hold fixed.

No coin flip, no state line, no event. That is exactly why the argument collapses into *which controls to include* — and why reasonable people disagree.

What controls try to do

Without a design, the fallback is **adjusting**: compare men and women who are *alike on paper*.

Same education, same experience, same industry, same occupation, same hours — then measure the gap **that remains**.

The hope: the remaining gap compares **like with like**, so what is left is closer to unequal pay for equal work.

Controls are a substitute for the missing design —
a comparison group built by bookkeeping instead of by chance.

Plug the numbers in

Blau and Kahn (2017), U.S. panel data (PSID), full-time workers, 2010:

comparison	women's pay ratio	remaining gap
raw (no controls)	79.3%	$100 - 79.3 = \mathbf{20.7}$
+ education, experience	82.1%	$100 - 82.1 = \mathbf{17.9}$
+ industry, occupation, union	91.6%	$100 - 91.6 = \mathbf{8.4}$

Controls shrink the gap by $20.7 - 8.4 = \mathbf{12.3}$ points — and education barely matters ($20.7 - 17.9 = 2.8$): **where people work** does the shrinking.

Blau and Kahn (2017), *Journal of Economic Literature* 55(3), 789–865.

The trap: some controls are outcomes

“Controlling for occupation” treats occupation like age — a fixed background fact. But **occupation is partly an outcome** of the process being measured:

- if hiring or promotion **steers women away** from high-paying occupations,
- or if **anticipated** child-rearing shapes career choices years ahead,

then part of the gap *runs through* occupation.

Controlling for an outcome deletes part of the effect you are measuring.
The 8.4 is not “the true gap” — some of the effect was controlled away.

Neither extreme is right

position	reading	error
control for nothing	“the gap is 20.7”	attributes <i>every</i> difference to discrimination
control for everything	“the gap is 8.4”	controls away effects that <i>run through</i> occupation

The honest reading: the like-for-like gap lies **somewhere between 8.4 and 20.7** — and **the data alone cannot say where**, because there is no design.

This is an identification problem, not (yet) an answer.

What would a design need?

The raw comparison fails for want of three things:

1. an **event** — datable, so there is a before and an after;
2. a **comparison path** — a group tracing the counterfactual;
3. **pre-trends** — history to test the comparison against.

Within the gap, there is one place that has all three.
That is the final part.

Active Learning 7

A wage comparison between men and women proposes five controls. For each: is it **background** (safe to control) or **partly an outcome** of the process being measured (controlling it deletes effect)? **One sentence of justification each — the sentence is the marks.**

- (a) age
- (b) education completed at age 22
- (c) current occupation
- (d) hours worked last year
- (e) whether the worker has children

Four minutes. Paper and pencil.

PART 8

The Child Penalty: An Event Study

The birth of a first child is what the raw gender comparison never had:

- **datable** — a before and an after, to the month;
- it happens to **identifiable people**, whose earnings run for years on both sides;
- a **comparison path** is at hand: fathers' earnings across the same event.

Event study. Difference-in-differences stretched over many periods: track treated and comparison outcomes for years around the event, not just one before and one after.

The study

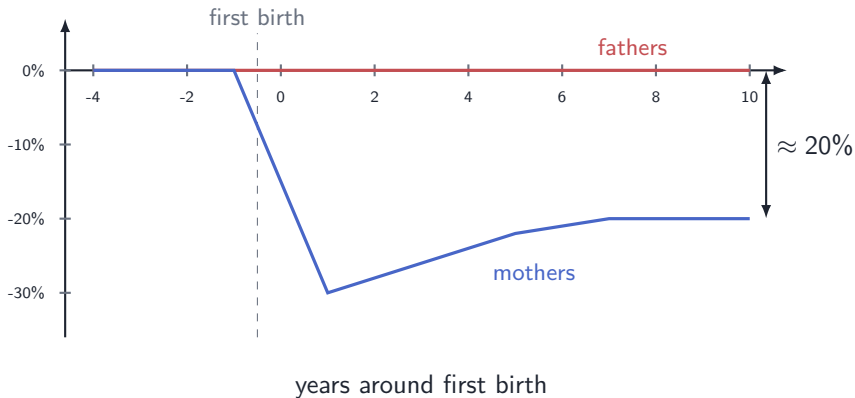
Kleven, Landais and Sørensen (2019): **Danish administrative data** — every parent's earnings, every year, linked across decades.

Line up mothers and fathers by **event time**: years relative to first birth. Compare each parent's earnings with their **own pre-birth path**, and mothers against fathers.

Denmark: subsidized childcare, long parental leave — a *hard* case for finding penalties, not an easy one.

Kleven, Landais and Sørensen (2019), *American Economic Journal: Applied Economics* 11(4), 181–209.

Draw the graph



Stylized magnitudes from Kleven, Landais and Sørensen (2019): earnings relative to own pre-birth path.

What it finds

- Mothers' earnings drop $\approx$ **30%** in the years around first birth, and settle $\approx$ **20%** below their pre-birth path — **long run**.
- Fathers' earnings are **essentially unchanged** by the same event.
- The drop splits into hours, participation, and wage rates — all three move.

In Denmark, the share of remaining gender earnings inequality accounted for by children has grown to **roughly 80%**. Cross-country estimates of the long-run penalty reach $\approx$ **31% in the U.S.**

Kleven et al. (2019), *AEA Papers & Proceedings* 109, 122–126.

Part 7 posed the problem. This is the answer.

- Part 7's comparison had **no event, no counterfactual path, no pre-trends** — so it collapsed into a fight over controls.
- The child penalty has **all three**: a dated event, the father's path, and years of flat pre-trends.

It does not settle *why* — preference, constraint, norm — and it measures the penalty, not every part of the gap.

But it converts “argue about controls” into “read the design” —
for the largest single component of the gap.

The whole argument on one frame

1. A **correlation** is a fact; a **causal claim** is about a **counterfactual** — which is never observed.
2. A **control group** is a manufactured counterfactual; randomization builds the best ones, and economists rarely get it.
3. **DiD**: two subtractions kill the permanent gap and the shared trend — *if* trends were parallel.
4. **Pre-trends** are the test; without them, an estimate is a hope.
5. Real disputes — minimum wage, gender gap — are fights over **the comparison and the data**. That is the discipline working.

Active Learning 8

Average annual earnings around first birth (invented):

	2 yrs before	2 yrs after
mothers	\$50,000	\$38,000
fathers	\$56,000	\$58,000

- (a) Compute the DiD penalty **both routes**; express it as a share of mothers' pre-birth earnings.
- (b) What must be true of **fathers** for this to be causal?
- (c) One concrete way (b) could **fail** — and the *direction* it pushes the estimate. **The reasoning is the marks.**

Five minutes. Paper and pencil.

A causal claim is a claim about something that never happened.

The whole craft of empirical economics is building —
and defending — the counterfactual.